Edward Ajayi

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Education

Carnegie Mellon University
Master of Science in Engineering Artificial Intelligence — *Aug. 2024 – May. 2026*

- **Relevant Courses:** Natural Language Processing, Deep Learning, Question Answering with LLMs
- **Specializations:** Data Science, Machine Learning, NLP
- **GPA:** 3.73/4.0

Ekiti State University
Bachelor of Science in Computer Science — *Oct. 2014 – Nov. 2019*

- **Specializations:** Computer Science
- **GPA:** 4.17/5.0

Research Interests

Human-Centered AI, Socially Intelligent AI, Multi-Agent Systems, LLM Reasoning & Evaluation, AI Alignment, Cognitive Modeling.

Selected Publications

1. **E. Ajayi** and P. Mitra. *HumorGen: Cognitive Synergy for LLM Humor Generation via Persona-Based Distillation. Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026, Main Conference).* [ACL Anthology]
2. **E. Ajayi**, J. Karangwa, B. Manyara, and P. Mitra. *SIDE-Bench: Evaluating Social Interaction and Conversational-Partner Intent in Multi-Agent Environments.* Under Review at *ICLR 2027*. [PDF]
3. **E. Ajayi** and P. Mitra. *HumorRank: An Automated Leaderboard for Evaluating Humor Generation in Large Language Models.* Under Review at *ACL Rolling Review (ARR)*, 2026. [arXiv]
4. **E. Ajayi** and P. Mitra. *Cross-Lingual Cognitive Synergy for Constrained Humor Generation in LLMs. Working Notes of the Conference and Labs of the Evaluation Forum (CLEF 2026), CEUR-WS.* **Best Paper, JOKER Lab**. [CEUR-WS]
5. **E. Ajayi**, M. Kachweka, M. Deku, and E. Aiken. *A Machine Learning Approach for Detection of Mental Health Conditions and Cyberbullying from Social Media. Proceedings of Machine Learning Research (PMLR), AAAI-26 AI in Medicine and Healthcare Bridge.* **Best Paper Award**. 2026. [PMLR]
6. **E. Ajayi**, B. Tadele, E. Umwari, M. Deku, P. Singadi, C. Edeh, and J. Udahemuka. *Language Diversity: Evaluating AI Performance on African Languages. Deep Learning Indaba 2026 Research Track, IJCAI 2026 Proceedings, to appear.* [arXiv]
7. **E. Ajayi**, M. Alaba, D. Stephen, and P. Bakare. *AfriEconQA: A Benchmark for Quantitative and Temporal Reasoning over World Bank Economic Reports.* Under Review at *ACL Rolling Review (ARR)*, 2026. [arXiv]

Experience

Research Associate
Spatial and Language Technologies Lab, Carnegie Mellon University — *Jun. 2026 – Present*

- Building efficient multimodal humor generation models using LLMs and vision-language models (VLMs).
- Developing evaluation benchmarks for Large Language Models on subjective generation tasks and alignment.
- Conducting research on computational humor, multilingual NLP, and reasoning abilities of foundation models.

AI Explorers Fellow (Supervisor: Prof. Rada Mihalcea)
Michigan Artificial Intelligence Lab (Michigan AI), University of Michigan — *Jul. 2026 – Present*

- Conducting research on the effect of group membership and social identity on agent behavior in multi-agent systems.

Predoctoral Research Fellow
Human-Algorithm Interaction Lab, Oxford Saïd Business School, University of Oxford, UK — *Feb. 2026 – May. 2026*

- Conduct research on Human-AI collaboration, decision-making, and AI adoption in service systems at the Human-Algorithm Interaction Lab.

Research Assistant
Spatial and Language Technologies Lab, Carnegie Mellon University — *Aug. 2025 – May. 2026*

- Conducting research on NLP for automatic humour detection and generation, dataset preparation, model training, evaluation, and benchmarking.
- Contributing to multilingual and low-resource language technologies focused on culturally grounded language models.

Research Intern - Business Automation
Carnegie Mellon University — *Jun. 2025 – Aug. 2025*

- Built an agentic AI workflow framework using LangGraph and FastAPI to automate complex business workflows.
- Researching and implementing AI safety protocols and robust LLM reasoning for ethical alignment and deployment.

- Analyzing agent performance to establish benchmarks for human-in-the-loop oversight and develop explainability metrics to support transparent corporate accountability.

Client Service Officer
Stanbic IBTC Bank *Dec. 2021 – Jul. 2024*
- Leveraged data-driven insights to cross-sell over N400M in assets, enhancing client trust and digital adoption by 25%.
- Managed client relationships, identifying service gaps to improve operational efficiency.

Technical Skills

Languages: Python, SQL, Bash
Generative AI & LLMs: LangGraph, OpenAI Agents SDK, RAG, LLM Fine-Tuning, Prompt Engineering, DPO
Deep Learning & ML: PyTorch, Hugging Face Transformers, Scikit-Learn, NumPy, Pandas, Matplotlib, Seaborn
Natural Language Processing: spaCy, NLTK, Sentiment Analysis, Text Processing
Data Engineering & Ops: Docker, AWS, Azure (Certified Data Scientist Associate), FastAPI, Neo4j, BeautifulSoup, Jupyter

Selected Projects

HumorRank — LLM Evaluation & Ranking Framework | *Python, Evaluation* Leaderboard [Paper]
- Developed a scalable evaluation framework for ranking outputs from multiple LLMs on subjective generation tasks.
- Implemented pairwise comparison with probabilistic ranking using the Bradley-Terry model.
- Designed a candidate generation → evaluation → leaderboard workflow for scalable benchmarking.

HumorGen — Open-Source LLM Humor Generation Suite | *PyTorch* Website Hugging Face
- Developed HumorGen, releasing 14 open-source language models for controllable humor generation.
- Implemented alignment pipelines using supervised fine-tuning and Direct Preference Optimization (DPO).
- Designed the Cognitive Synergy framework for diverse humor generation.

MyTorch — Minimal Deep Learning Framework Built from Scratch | *Python*
- Implemented core tensor operations, automatic differentiation, activation layers, and SGD without deep learning libraries.
- Designed a PyTorch-style API enabling end-to-end model training while illustrating neural network internals.
- Built modular components for educational experimentation with deep learning systems.

Mental Health & Cyberbullying Detection System (AAAI Oral) | *Transformers* GitHub [arXiv]
- Built an end-to-end multi-label classification system for detecting mental health signals and cyberbullying from social media
- Combined classical ML methods with transformer embeddings for robust prediction on noisy datasets.
- Produced results leading to an Oral paper acceptance at AAAI-26 AIMedHealth.

Training & Conferences

Oral & Poster Presentation, AAAI Conference on Artificial Intelligence Singapore
AAAI *Jan. 2026*
- Presented my paper on ML for mental health and cyberbullying detection from social media, and engaged with academics and researchers at the conference.

Participant, Technical AI Safety Course Online
Bluedot Impact *May 2025 – Present*
- Training in technical AI safety, focusing on alignment approaches, mechanistic interpretability, and catastrophic risk mitigation.

Participant, Digital Humanities Summer School Rwanda
DIGIHUM/CMU-Africa *Jul. 2025*
- Engaged in computational methods for humanities research, including NLP applications.

Volunteering/Leadership

Founder & Organizer, CMU-Africa NLP Group
Carnegie Mellon University Africa *Apr. 2025 – Present*
- Managing the group activities and overseeing research projects focused on language technologies.

President, CMU-Africa Research Club
Carnegie Mellon University *Jan. 2025 – Dec. 2025*
- Led 12 major events in 2025 supporting student research.
- Launched website with research guides, conference updates, and collaboration hub.
- Organized first Student Research Showcase with 20+ posters and workshops.
- Collaborated with ACM SIGCOMM and Upanzi Network for remote conference participation on campus.
- Built an active research ecosystem that provided critical peer support, contributing to over 20 student acceptances to top-tier venues including NeurIPS and AAAI in 2025.

Reviewer, AAAI AIMedHealth Bridge 2026
AI for Medicine and Healthcare *Nov. 2025 – Nov 2025*
- Reviewed submissions in applied machine learning and AI for healthcare under the AAAI AIMedHealth Bridge 2026 program.

Honors and Awards

- **Winner**, CLEF JOKER 2026 Task 4 — Best-performing humor generation system in English, Spanish, and French.
- Mastercard Foundation Scholarship, Carnegie Mellon University Africa
- Federal Scholarship Board Award for Undergraduates, Ekiti State University